Fig. 1 Comparison of Wide-Angle Crop (Digital Zoom), Super-Resolution, and true Optical Zoom active acquisition.

Dual-Camera Active Acquisition for Automated Small-Object Dataset Construction

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Deep learning performance on small objects is frequently bottlenecked by the quality and quantity of training data rather than model architecture. We present a dual-camera active acquisition system that automates small-object dataset generation by pairing a fixed wide-angle camera for persistent detection with a Pan-Tilt-Zoom (PTZ) camera for optical verification. The system runs concurrent YOLO detector instances on an embedded NVIDIA Jetson Orin AGX platform, achieving 35.9 FPS on the GPU for wide-area search and 21.5 FPS on the DLA for PTZ verification. A predictive control framework compensates for mechanical and pipeline latencies to transition from open-loop slewing to closed-loop PID tracking. We demonstrate the system on outdoor drone tracking, where the PTZ autonomously acquires and maintains lock on a small aerial target across a 42-second flight. The active acquisition paradigm replaces digital zoom and super-resolution with true optical zoom, providing ground-truth-quality imagery of targets that occupy fewer than 15×15 pixels in the wide field.

I. Introduction

While recent surveys [1, 4] have demonstrated significant architectural improvements in generic object detection, they have not fully solved a problem that has persisted for decades: small object detection. The core issue remains one of pixel-level information loss, since no amount of digital zoom can recover details that were never sampled. Although some super-resolution algorithms show promise by recovering detail via temporal information, these methods are often computationally intensive and ill-suited for strict real-time inference [2].

Even with high-resolution imagery, real-time detection pipelines typically downsample inputs to manageable resolutions (e.g., 640×640) to satisfy compute constraints. This reduction inherently compresses distant targets into featureless blobs (Fig. 1) that are difficult to classify without relying on temporal context (as humans do) or additional sensor modalities [3].

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Figure 1 illustrates the challenge in distant aerial surveillance, where a target may occupy fewer than 15×15 pixels in a wide-angle feed. At this resolution, distractors (i.e. birds, cloud edges, specular highlights, or sensor noise) are indistinguishable from the target of interest. Standard digital zooming (cropping) only magnifies these ambiguities [5]. To achieve robust detection, a dataset must contain examples where these confusing cases are resolved, yet human labelers often cannot distinguish them in raw wide-angle footage.

We address this by replacing the human labeler with a dual-camera system that automates the construction of small-object datasets. A fixed wide-angle camera provides persistent coverage, while a controllable Pan-Tilt-Zoom (PTZ) camera provides resolution on demand. The key technical challenge lies in the handover: the system must move a mechanical lens to capture a moving target based on delayed visual data, compensating for the combined pipeline and mechanical latency.

The contributions of this work are: (i) a hardware-software architecture that synchronizes wide-angle search with narrow-angle verification on an embedded edge platform; (ii) a predictive control formulation that compensates for system latency to enable reliable active target acquisition from wide-angle cues; and (iii) experimental characterization of detector throughput, GPU/DLA parallel inference, and a drone tracking demonstration validating end-to-end system operation.

II. Related Work

A. Small Object Detection and Resolution Limits

The fundamental bottleneck in small object detection is the scarcity of distinguishing features. When a target occupies few pixels (i.e. 15×15 pixels), class-defining details are often lost entirely [1]. Classical solutions involving super-resolution (SR) attempt to reconstruct this lost detail [2].

However, reliance on SR for scientific data collection is problematic. First, SR is fundamentally generative; it estimates high-frequency details based on learned priors, creating a risk of hallucination where the model reinforces its own biases [5]. Second, high-fidelity generative models required for scientific validity often require > 300 ms per frame on edge accelerators [18], whereas lightweight models sacrifice reconstruction quality. Our system achieves mechanical slew-and-settle in a comparable time window while obtaining true optical ground truth rather than hallucinated details.

B. Active Acquisition vs. Continuous Tracking

Most PTZ tracking literature focuses on the control problem of keeping a target centered in the frame [7, 8]. This requires mitigating total system latency (video encoding + network + mechanical response), which for IP-based systems frequently ranges from 200–500 ms [17]. Our work addresses active acquisition, or "slew-to-classification." Unlike continuous tracking, where the objective is persistence, our objective is information gain via discrete spot-checks.

Existing active perception systems like VIGIA-E [11] typically optimize for broad area coverage or anomaly detection. Our system instead functions as a sparse query mechanism: it identifies specific low-confidence candidates in the wide field and commits the PTZ resource to verifying them individually.

C. Sensor-Driven Labeling

Reducing manual annotation is a central goal of both semi-supervised learning and active learning. Pseudo-labeling methods such as ASTOD [12] attempt to retrain models using high-confidence predictions, but this approach often fails in the small-object regime where the detector is consistently uncertain [13]. Active learning strategies like PPAL [15] identify informative samples but still require a human loop [14].

Our active acquisition creates a fully automated hybrid: the selection logic of active learning (targeting less confident samples) is paired with PTZ verification instead of a human annotator. While the target is ambiguous at 15×15 pixels, the zoomed view restores it to a regime ($> 100 \times 100$ pixels) where off-the-shelf detectors achieve near-perfect accuracy [6]. By physically bridging the gap between the surveillance view and the high-resolution training distribution, we convert a difficult small-object inference problem into a trivial classification task.

Fig. 2 Physical hardware configuration. Two FLIR M300C cameras are mounted on a shared three-foot stand with ≈ 0.5 m baseline separation. The left unit serves as the fixed wide-angle camera; the right unit operates as the PTZ camera.

III. System Overview and Formulation

A. Hardware and Software Roles

The system consists of two FLIR M300C cameras mounted on a shared three-foot stand with a fixed relative pose (Fig. 2): (1) a fixed wide-angle camera providing continuous coverage and running a real-time YOLO-family detector on the primary GPU, and (2) a PTZ camera whose pan, tilt, and zoom are commanded via an HTTP control interface. Both cameras are powered by a 12.8 V 50 Ah LiFePO4 battery, selected because the dual-camera current draw exceeded the capacity of a standard DC power supply unit. The cameras connect to an NVIDIA Jetson Orin AGX (JetPack 4.7, CUDA 12.6) via a PCIe Ethernet network card with IEEE 802.3at PoE detection, confirmed safe for the non-PoE FLIR devices. The baseline separation $b \approx 0.5$ m produces a parallax angle of $\arctan(0.5/100) \approx 0.29°$ at 100 m range, which is negligible for the initial open-loop slew command.

The software architecture is organized in three layers (Fig. 5). The first layer (L1) provides a Kivy-based GUI with a streamer thread displaying dual camera feeds and exposing controls for zoom, movement speed, tracking mode, and PID tuning parameters. The second layer (L2) runs six threads: a capture thread, a YOLO processing thread, and a draw thread for each camera. The third layer (L3) implements the tracker process, which operates in two phases: Phase 1 performs target acquisition from the stationary camera, and Phase 2 transitions to active tracking on the PTZ camera via a coordinate mapper, centering controller, zoom controller, PID controller, and HTTP command interface. If the PTZ loses detection during Phase 2, the system falls back to stationary camera tracking; once the PTZ reacquires the target, it resumes PTZ-based control.

B. Problem Formulation

We formulate the active acquisition as a greedy utility maximization problem. Let $I^w_{t_{cap}}$ be the wide-angle frame captured at time t_{cap}, and let the PTZ state be defined by telemetry $\tau_t = (\theta, \phi, z, \dot{\theta}, \dot{\phi})_t$, explicitly including angular velocity to model inertia. The wide detector produces candidate detections $D_{t_{cap}} = \{(b^w_i, s^w_i, c_i)\}_i$.

The system must select an action a_t at decision time t that will take effect at a future arrival time $t_{arr} = t + \delta$, where δ accounts for processing and mechanical slew delays. The objective is to maximize the expected utility of the next captured observation:

$$a^\star = \arg\max_{a \in \mathcal{A}} \mathbb{E}\left[U(I^p_{t_{arr}}, \tau_{t_{arr}}) \mid D_{t_{cap}}, \tau_t\right] \tag{1}$$

This optimization is subject to operational constraints: the PTZ must be available (not currently slewing) and the target

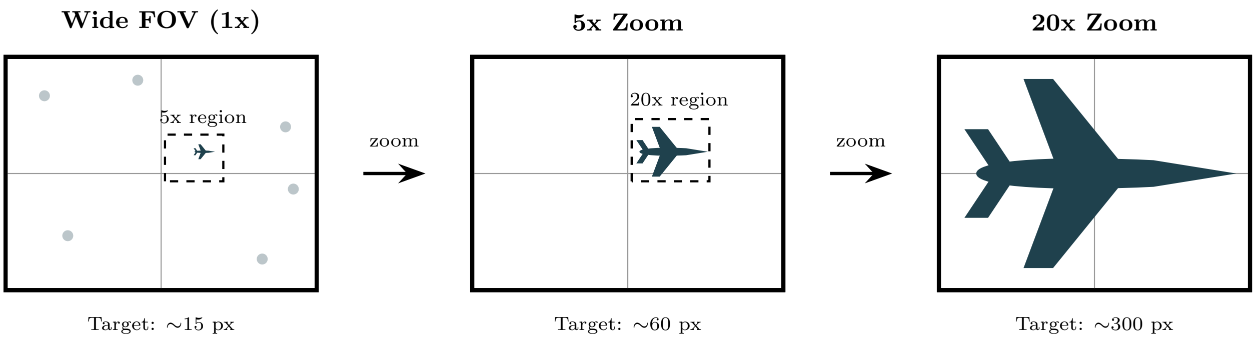

Fig. 3 Visual progression of an active acquisition sequence. A 15-pixel target in the wide field (1x) is indistinguishable from noise. The system cues the PTZ to an intermediate 5x zoom for acquisition, then a 20x zoom for final detailed verification, revealing the aircraft structure clearly.

must be within the reachable field of view. The utility function $U(\cdot)$ prioritizes high-confidence verification, localization precision, and sample diversity.

IV. Calibration and Cross-View Geometry

A. Camera Model

Both cameras are modeled with intrinsics (K_w, K_p) and distortion parameters. Let $\Pi(\cdot)$ be perspective projection with distortion, and let R_{pw}, t_{pw} map 3D points from wide-camera coordinates to PTZ-camera coordinates.

For sky targets at long range, parallax between cameras is negligible when the baseline is small relative to range. With a camera baseline of $b = 0.5\,\mathrm{m}$ and a minimum operating range of $R = 100\,\mathrm{m}$, the angular parallax is $\arctan(b/R) = \arctan(0.5/100) \approx 0.29°$. This is well within the PTZ field of view at any zoom level, so a direction-only transfer provides a sufficiently accurate initial seed for the PTZ to acquire the target. Once the PTZ is slewed to this seed location, its onboard detector refines the tracking lock. In this regime, ray directions on the unit sphere are used for cross-view mapping rather than estimated depth.

B. Mapping Wide Detections to PTZ Pan/Tilt Commands

Let (u, v) be the center of a wide detection box b^w in pixel coordinates. After undistortion, the bearing direction in the wide camera frame is

$$\hat{d}_w = \frac{K_w^{-1}[u\ v\ 1]^\top}{\|K_w^{-1}[u\ v\ 1]^\top\|}. \tag{2}$$

Transforming to the PTZ base frame gives $\hat{d}_p = R_{pw}\hat{d}_w$. Using a conventional yaw–pitch parameterization, the commanded pan (yaw) and tilt (pitch) are

$$\theta = \operatorname{atan2}(\hat{d}_{p,y}, \hat{d}_{p,x}), \quad \phi = \operatorname{atan2}\left(\hat{d}_{p,z}, \sqrt{\hat{d}_{p,x}^2 + \hat{d}_{p,y}^2}\right). \tag{3}$$

Because PTZ motion and video pipelines have latency, $\hat{d}_p$ is preferably computed from a short-horizon prediction of target motion rather than the instantaneous detection center.

C. Zoom Selection by Target Angular Size

Let h^w be the detected box height in wide pixels. Under small-angle approximation, the apparent angular height is approximately $\alpha \approx h^w/f_w$, where f_w is the wide focal length in pixels. To obtain a desired PTZ pixel height h^p_{des}, select a PTZ focal length $f_p(z)$ such that

$$h^p_{\text{des}} \approx \alpha f_p(z) \approx \frac{h^w}{f_w} f_p(z), \tag{4}$$

then choose the zoom z whose calibrated $f_p(z)$ best matches the required value, clipped to PTZ limits. This approach avoids explicit range estimation and is well suited for distant aircraft.

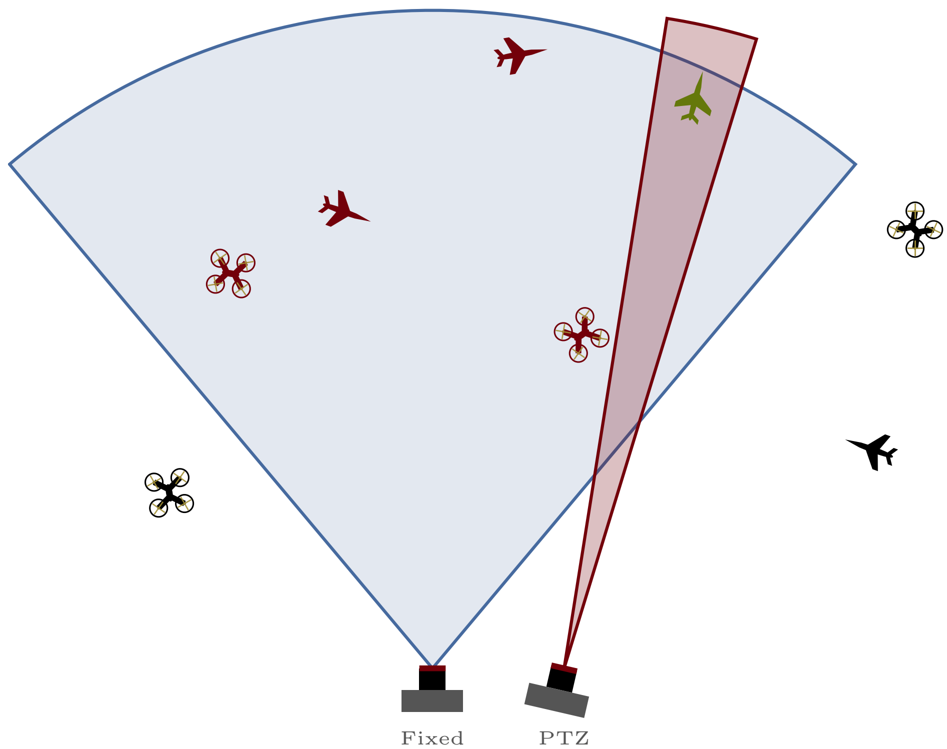

Fig. 4 Field of view comparison. The wide camera (blue) covers approximately 60° horizontally for persistent surveillance, while the PTZ camera (red) provides a narrow, steerable view.

V. PTZ Triggering, Tracking, and Scheduling

PTZ tracking is a coupled perception-and-control problem with dynamic imaging conditions and control delays. Prior PTZ tracking evaluations emphasize that camera motion, latency, and re-centering quality dominate performance differences in practice.

We use wide-camera tracklets to provide temporal coherence and to predict target bearing during PTZ motion. A lightweight predictor (e.g., constant angular velocity with α–β filtering or a Kalman filter) provides a predicted bearing $\hat{d}_w(t + \Delta)$ given command latency Δ, consistent with classical pan/tilt tracking formulations.

A. Utility Function for Triggering and Redundancy Control

The trigger utility balances value and cost. A practical form is a weighted sum of terms:

$$U(o) = \lambda_1 \,\mathrm{Unc} + \lambda_2 \,\mathrm{SizeGain} + \lambda_3 \,\mathrm{Novelty} - \lambda_4 \,\mathrm{Cost}, \tag{5}$$

where Unc increases when the wide detector is unsure (so PTZ confirmation is valuable), SizeGain estimates how much the PTZ will increase target pixels, Novelty reduces redundancy by down-weighting near-duplicates, and Cost captures PTZ time-on-target, slew distance, and opportunity cost (only one PTZ target at a time). In the current implementation, the system triggers on the largest detected target with confidence above the acquisition threshold ($s > 0.40$), corresponding to a simplified policy where $\lambda_1 = 0$ and the system always acquires available targets.

VI. Automated Dataset Construction

A. Label Sources and Cross-View Transfer

The PTZ view is treated as a higher-fidelity label source when it produces a confirmed detection for the target class. The label transfer module maps the PTZ detection back into the wide image coordinate system.

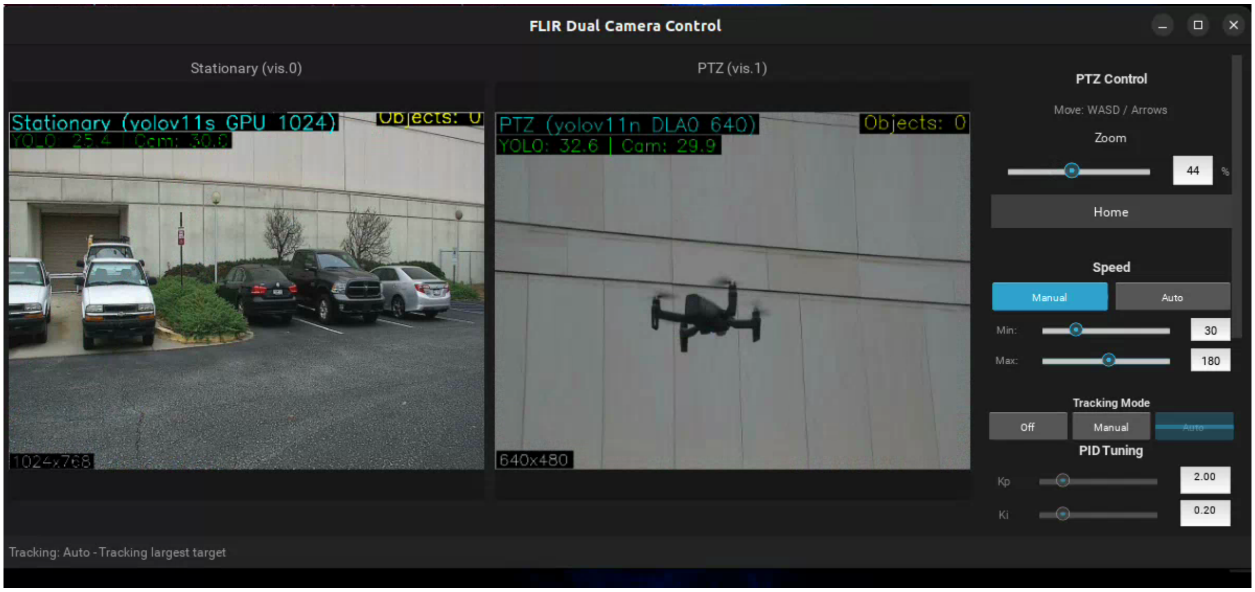

Fig. 5 Live screenshot of the three-layer software interface during a drone tracking test. The stationary camera (left) runs YOLOv11s on the GPU at 1024×1024 resolution for wide-area detection. The PTZ camera (center) runs YOLOv11n on the DLA at 640×640, here tracking the drone at optical zoom. The control panel (right) exposes PTZ zoom, movement speed, tracking mode selection, and PID tuning parameters.

Algorithm 1 Real-time wide-to-PTZ acquisition

Init f_θ (wide), $\mathcal{T}$ (tracker), g (PTZ), $\mathcal{D}$ (data)
for each wide frame I_t^w **do**
    $D_t \leftarrow f_\theta(I_t^w)$; $\mathcal{T} \leftarrow \text{UpdateTracks}(\mathcal{T}, D_t)$
    Compute $U(o)$ for tracks (conf, size, novelty)
    Select $o^\star \leftarrow \arg\max U(o)$ s.t. availability
    if $U(o^\star) > \tau_{\text{trigger}}$ **then**
        Predict $\hat{d}_w(t+\Delta)$; map to (θ, ϕ, z)
        Command PTZ: $g(\theta, \phi, z)$; get $I_{t'}^p$, $\tau_{t'}$
        Run PTZ detector: $D_{t'}^p \leftarrow f_{\theta_p}(I_{t'}^p)$
        if Quality OK AND max $s^p > \tau_{\text{confirm}}$ **then**
            $b^{w \leftarrow p} \leftarrow \text{Project}(b^p, \tau_{t'})$
            Commit $(I_t^w, b^{w \leftarrow p}, c)$ to $\mathcal{D}$
        end if
    end if
end for
Periodically retrain f_θ on $\mathcal{D}$

For distant targets, direction-only transfer proceeds by converting the PTZ bounding box corners into bearing rays, rotating those rays into the wide camera frame, and projecting them into the wide image plane. Let (u_k^p, v_k^p) be PTZ box corners; compute PTZ rays $\hat{d}_k^p$ from K_p^{-1}, rotate to wide frame $\hat{d}_k^w = R_{wp}\hat{d}_k^p$, then project:

$$[u_k^w, v_k^w, 1]^\top \propto K_w \hat{d}_k^w, \quad k \in \{1, 2, 3, 4\}. \tag{6}$$

The transferred wide box is the tight axis-aligned rectangle enclosing the projected corners.

B. Quality Gates and Drift Prevention

Self-training and pseudo-labeling can suffer from confirmation bias if low-quality pseudo-labels are admitted. In this system, the primary drift control mechanism is the physical zoom verification: many ambiguous wide detections

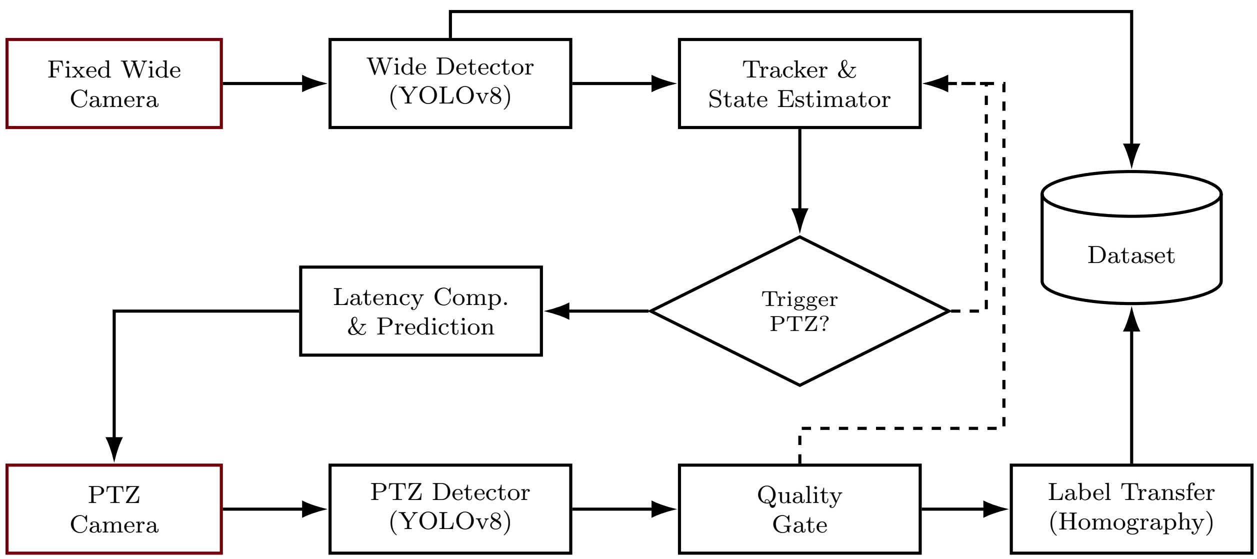

Fig. 6 Automated label-generation pipeline. Wide-field tracking selects candidates, PTZ verifies optically, and homography maps labels to the wide view.

become unambiguous at higher resolution. Remaining failure modes are handled by quality gates that reject samples when motion blur is excessive, exposure is saturated (e.g., sun glare), the target is at the frame boundary, or the PTZ detector disagrees with the wide detector class. Each gate is implemented as a deterministic predicate so that dataset inclusion is reproducible and auditable.

C. Dataset Schema

Each committed example stores the wide frame (or short clip), the transferred label, and acquisition metadata. Table 1 defines the record schema.

VII. Experimental Validation

A. System Configuration

The complete system runs on an NVIDIA Jetson Orin AGX with JetPack 4.7 and CUDA 12.6. Table 2 summarizes the hardware and software configuration. The stationary wide camera provides a $60° \times 45°$ field of view for persistent surveillance, while the PTZ camera operates at configurable zoom levels of 25%, 35%, and 45% during tracking.

B. Detector Throughput

The two detector instances are split across compute units to avoid GPU time-slicing overhead. The stationary camera runs YOLOv11s on the GPU, where larger model capacity aids feature extraction on distant, low-resolution targets.

Table 1 Dataset record schema for each accepted acquisition.

Field	Description	Field	Description
timestamp_w	Wide-frame timestamp	timestamp_p	PTZ-frame timestamp
image_w	Wide image or clip key	telemetry_p	PTZ pan/tilt/zoom/exposure
bbox_w	Label box (wide coords)	bbox_p, score_p	PTZ box and confidence
class	Target class	quality_metrics	Blur, saturation, occlusion
score_w	Wide detector confidence	site_meta	Location, orientation, weather

Table 2 System configuration and measured detector throughput. All models use 640×640 input with FP16 precision.

Hardware & Software		
Compute platform	NVIDIA Jetson Orin AGX	
Software stack	JetPack 4.7, CUDA 12.6	
Cameras	2× FLIR M300C	
Power	12.8 V 50 Ah LiFePO4	
Network	PCIe Ethernet, IEEE 802.3at PoE	
Camera latency	0.2–0.6 ms (network)	
Wide FOV	60°H × 45°V	
Video codec	MJPEG (frame-independent)	
Detector Throughput		
	Stationary (GPU)	PTZ (DLA)
Model	YOLOv11s	YOLOv11n
Avg inference (ms)	27.9	46.5
Throughput (FPS)	35.9	21.5
Detection rate	100%	99%
Avg confidence	0.553	0.529

The PTZ camera runs YOLOv11n on the DLA, where zoomed-in targets are easier to classify. This split achieves true parallel inference with no resource contention.

The GPU delivers 2.2–2.8× higher throughput than the DLA per model, but running both models on the GPU causes time-slicing that degrades aggregate performance. The GPU/DLA split provides concurrent operation at 35.9 + 21.5 = 57.4 aggregate FPS with no contention (Table 2). The stationary detector achieves 100% detection rate with an average confidence of 0.553, while the PTZ detector achieves 99% detection rate with an average confidence of 0.529.

C. Tracking Controller

The PTZ centering loop uses a PID controller running at 30 Hz. Table 3 lists the controller parameters. Confidence thresholds are set asymmetrically: the acquisition threshold (0.40) is higher than the tracking threshold (0.35) to prevent the system from acquiring weak detections while allowing it to maintain lock through brief confidence dips. If the target is lost for more than 3.0 s, the system reverts to Phase 1 acquisition from the stationary camera.

Table 3 Tracking controller parameters.

Parameter	Value	Unit	Parameter	Value	Unit
PID gain K_p	1.5	—	Center deadzone	2	%
PID gain K_i	0.05	—	Position tolerance	0.2	deg
PID gain K_d	1.0	—	Lost threshold	3.0	s
Control rate	30	Hz	Acquire timeout	10	s
Acquire confidence	0.40	—	Reacquire timeout	5	s
Track confidence	0.35	—	Max PTZ speed	30	deg/s
Zoom levels	25, 35, 45	%	Min zoom scale	15	%

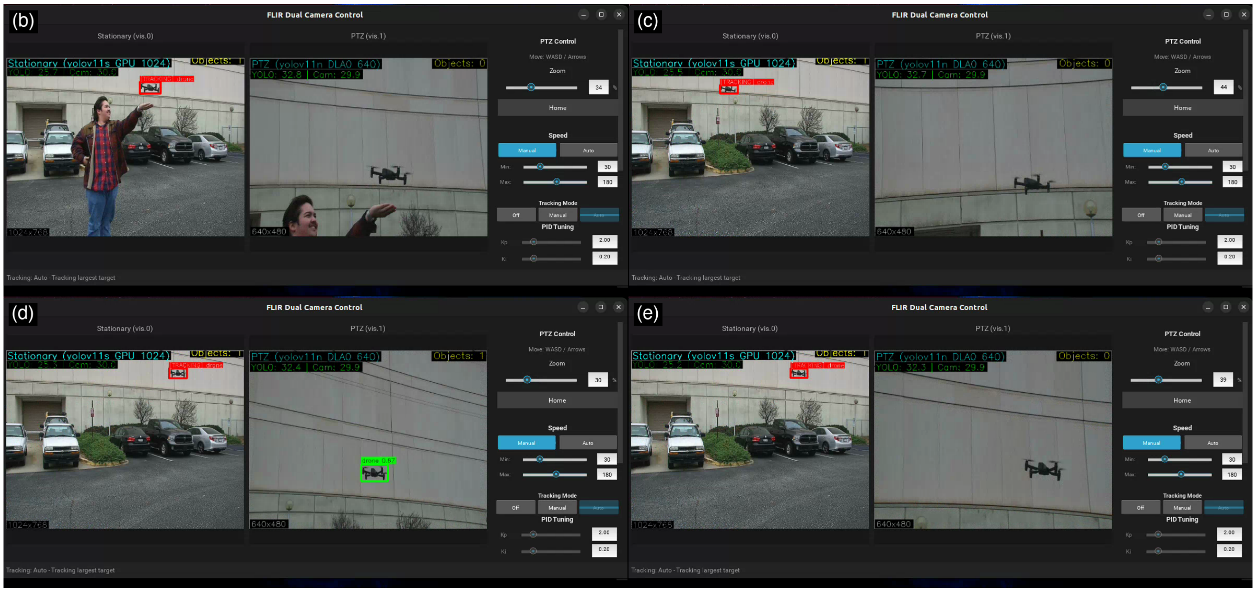

Fig. 7 Temporal montage from the 42-second drone tracking sequence. (a) Early tracking with PTZ slewing to target. (b) Active tracking at medium zoom with drone visible in both cameras. (c) PTZ detection at higher zoom with green bounding box on target. (d) Sustained tracking in final flight segment.

D. Video Codec Impact on Latency

Video codec selection has a direct impact on tracking latency. The system initially used H.264, which requires dependency frames (I/P/B) and introduces 2–4 frames of buffering latency before the decoder can deliver a complete image to the detector. Switching to MJPEG, which encodes each frame independently, eliminates this pipeline stall and provides near-instant decode. The reduction in end-to-end latency is critical for closed-loop tracking of fast-moving targets. Even with MJPEG latency reduction, the PID controller remains necessary to prevent overshoot when the PTZ tracks targets with high angular velocity.

E. Drone Tracking Demonstration

To validate end-to-end system operation, we conducted an outdoor drone tracking test. The system autonomously acquired and tracked a small commercial drone over a 42-second flight, producing approximately 42,000 frames across both cameras. Figure 7 presents a temporal montage of selected frames spanning the tracking sequence.

The stationary camera detected the drone at wide-angle resolution, providing bearing cues to the PTZ controller. The PTZ camera autonomously slewed to and tracked the target at optical zoom, maintaining centroid lock via PID control throughout the flight. The on-screen overlay (visible in Fig. 7) displays the model type, accelerator assignment, and inference resolution for each camera, confirming simultaneous GPU and DLA operation.

In the wide-angle view, the drone occupies approximately 20×15 pixels—insufficient for reliable classification. The PTZ view at 35–45% zoom resolves the target at approximately 120×90 pixels, a 6× increase in linear resolution that moves the target firmly into the regime where standard detectors achieve high accuracy.

F. In-Lab Airplane Detection

In-laboratory testing (Fig. 8) confirmed that the default YOLOv8n COCO "airplane" class detects large, high-resolution aircraft imagery displayed on a monitor but fails on small pixel footprints typical of the wide-angle surveillance view. This motivated training custom detection models on a drone detection dataset and on the Fine-Grained Visual Classification (FGVC) aircraft dataset, which provides ground-up RGB imagery of aircraft at a range of scales and orientations representative of the surveillance scenario.

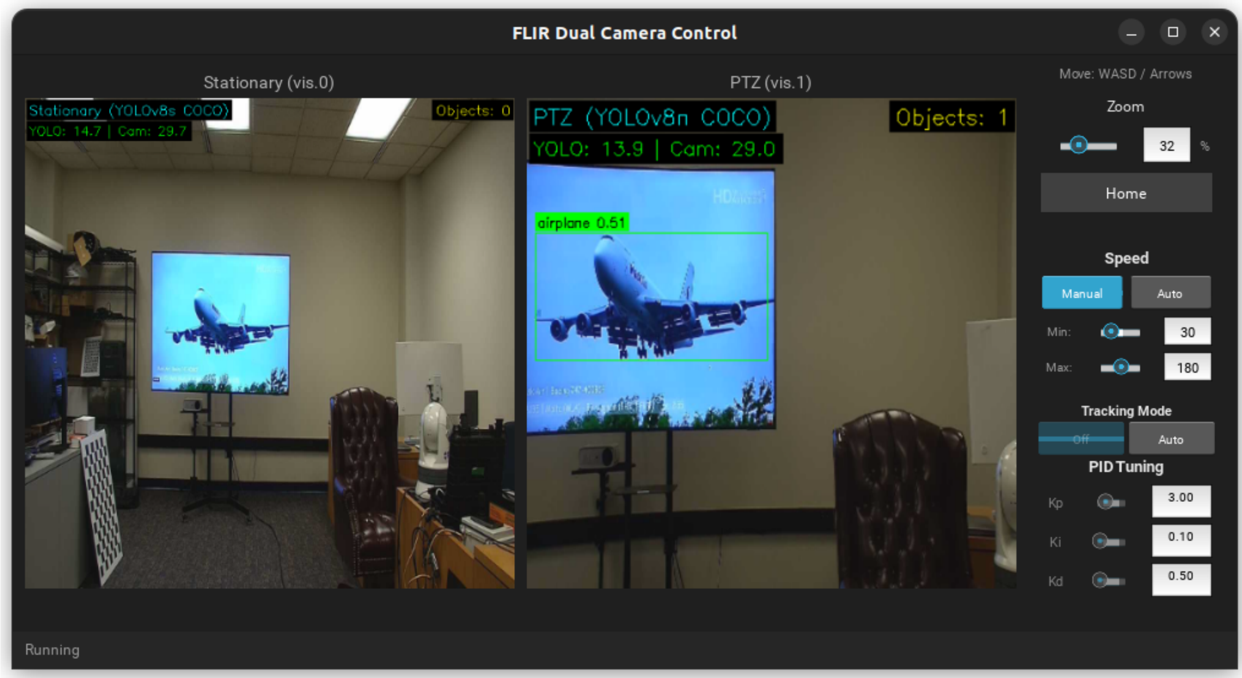

Fig. 8 In-laboratory validation. The PTZ camera detects and tracks an aircraft displayed on a monitor using YOLOv8n with COCO weights, confirming the detection pipeline and GUI control interface.

G. Airplanes Test Case

For the airplane surveillance application, the primary class is *airplane*. Hard negatives arise from birds, insects near the lens, distant drones, clouds with sharp edges, sensor noise, and specular highlights. The PTZ confirmation step naturally collects informative negatives: wide proposals that are rejected by PTZ as non-airplane are stored as hard negatives with contextual metadata.

Custom YOLO models trained on FGVC aircraft data are intended to replace the COCO-pretrained weights for the wide camera, targeting the small pixel sizes ($< 30 \times 30$) that COCO-trained models fail to detect. Benchmarking these custom models against default YOLO classes on a test set containing both long-distance and close-distance aircraft remains ongoing work.

VIII. Discussion

A. Telemetry Synchronization

Accurate label transfer requires time alignment between wide frames, PTZ frames, and PTZ telemetry. The system logs monotonic timestamps at capture and at inference and records the PTZ telemetry state at the exact time a PTZ frame is captured. If telemetry is sampled asynchronously, interpolation to the frame time reduces geometric error.

B. When PTZ Verification Helps Most

PTZ verification is most valuable when the wide detector frequently encounters ambiguous, small candidates whose pixel support is insufficient for confident classification. In such regimes, zoom provides additional evidence without changing the wide camera that will ultimately run the detector. This is especially beneficial when the deployment environment differs from training data and when pseudo-labeling alone would be unreliable.

C. Limitations

The system is constrained by single-PTZ availability and cannot zoom multiple targets simultaneously. Fast-moving targets may leave the PTZ field of view during slew, particularly when latency is high or the wide tracker is unstable.

Atmospheric turbulence and haze can reduce the effective benefit of zoom. Direction-only label transfer assumes small parallax; large baselines or nearer targets require depth-aware mapping. The current experimental characterization covers detector throughput and a single drone tracking scenario; larger-scale deployment with multiple target types and varying weather conditions is needed to fully assess operational robustness.

D. Future Work

The evaluation protocol for full deployment includes three baselines: a wide-only baseline without PTZ confirmation, a PTZ patrol baseline with scripted scanning, and a manual-zoom oracle on an audited subset. Primary metrics include label precision and recall, bounding-box IoU between transferred and audited labels, confidence uplift ($\Delta s = s^p - s^w$), and downstream small-object mAP after training on the constructed dataset. System metrics include slew-to-lock time, dwell time per target, reacquisition success rate, and PTZ availability contention. Ablation studies will isolate contributions from trigger thresholds, motion prediction, zoom scheduling, and label transfer methods. Multi-PTZ configurations can reduce contention and increase throughput via joint scheduling across targets.

E. Safety and Privacy

The airplane test case naturally focuses on sky regions; nonetheless, deployment constrains PTZ tilt limits and defines privacy-preserving regions to avoid capturing ground-level imagery. These constraints are implemented at the control layer by rejecting commands that enter prohibited zones.

IX. Conclusion

This paper presented a dual-camera active acquisition system for automated small-object dataset construction. A fixed wide-angle camera with real-time detection provides persistent coverage, while a PTZ camera delivers optical zoom for verification. The system runs on an NVIDIA Jetson Orin AGX, achieving 35.9 FPS (GPU) and 21.5 FPS (DLA) through a parallel GPU/DLA inference split that eliminates resource contention. PID-controlled PTZ tracking with MJPEG codec selection minimizes pipeline latency for closed-loop operation. We demonstrated end-to-end functionality through a 42-second outdoor drone tracking test, where the system autonomously acquired and maintained lock on a small aerial target, resolving it from ~20 pixels in the wide view to ~120 pixels under optical zoom. The active acquisition paradigm provides ground-truth-quality imagery that neither digital zoom nor super-resolution can match, enabling automated dataset construction for small-object regimes where manual annotation is impractical.

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